Electrostatics and Electric flux

Wed May 06 2026

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Vipul Deshmukh

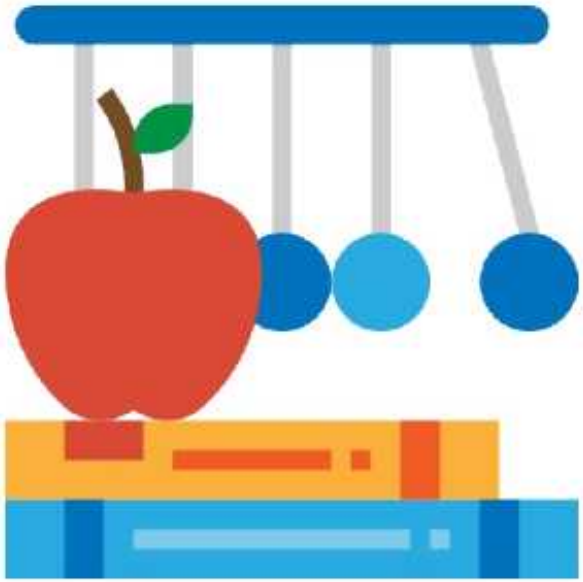
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Electrostatics

Grade 12 • Physics • Session 1

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Introduction

Can you think and tell us what is exactly happening in the image below? Have you seen this anywhere in movies or cartoons?



Introduction

Have you ever tried to deflect a stream of water with the help of a charged plastic scale?

Have you ever been curious about the reason behind it?



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Introduction



So, the word "electrostatics" means "study of charges when they are at rest".

or

Electrostatics deals with the study of forces, fields and potentials arising from static charges.

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History

- People noticed that there are certain pairs of objects which start attracting each other once they have been rubbed together. This attraction was not happening before the objects were rubbed together.
- Amber (called "electra" in Greek) was the first object for which people noticed the above behaviour. Other objects are Glass and silk, plastic and fur etc
- These observations were beginning of science of Electricity.

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History

- Researchers believed that rubbing objects with one another gives them an invisible property that enables them to attract each other.
- Benjamin Franklin gave this property the name of charge.
- Researchers were able to confirm that only two types of charges exist.
- Franklin chose the name "negative" and "positive" for these two types. He decreed that the charge that appears on glass rod on rubbing it with silk is positive.



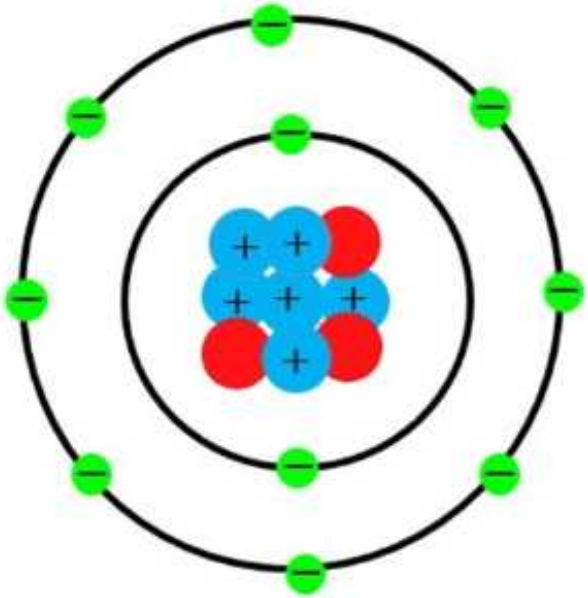
History

- When electron was discovered by J.J. Thomson, it was found that it possesses the same charge as possessed by a silk cloth on rubbing it with silk, hence electron was declared to have negative charge.

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Structure of an atom

$e = 1.6 \times 10^{-19} C$



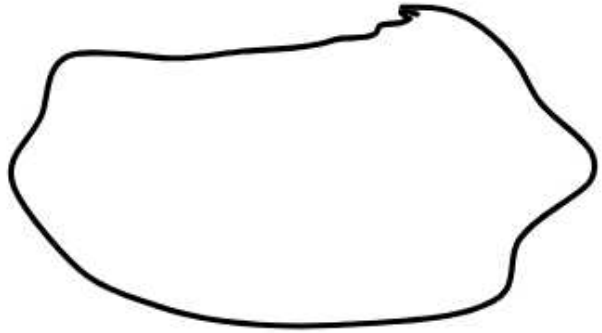
electron $(-e)$

protron $(+e)$

neutron

- We know, protons are bound in the nucleus (therefore immobile),but outermost shell electrons can move freely.
- Whenever objects acquire charge, electrons move from one object to another and not protons due to their immobility.

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Charge

10P = 10e → Q = 0

10P < 11e → Q = -e = -1.6×10^{-19} C

10P < 12e → Q = -2e = $-2 \times 1.6 \times 10^{-19}$ C

= -3.2×10^{-19} C

10P < 15e → Q = -5Xe

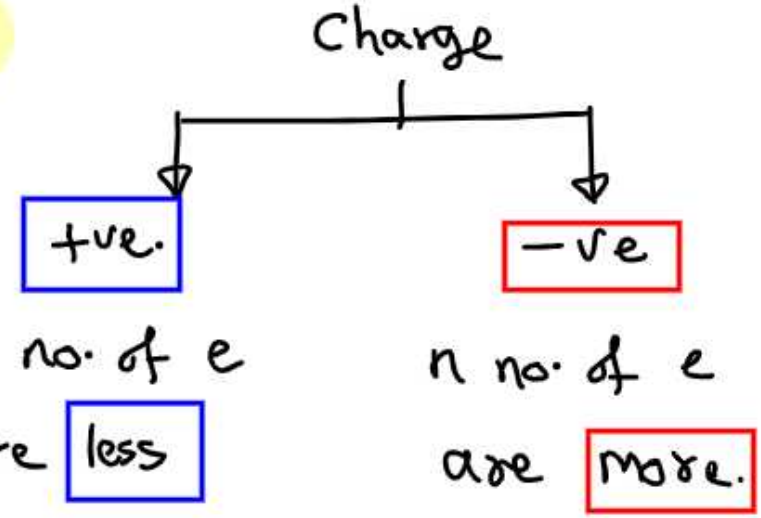
10P > 9e → Q = +e = $+1.6 \times 10^{-19}$ C

10P > 8e → Q = +2e = $+3.2 \times 10^{-19}$ C.

Quantization of charge.

$|Q| = ne$

$n \in I$



$Q = +8C$

$|Q| = ne$

$8 = n\,1.6 \times 10^{-19}$

$n = \dfrac{8}{1.6 \times 10^{-19}}$

$n = \dfrac{10}{16} \times 10^{19}$

$n = 5 \times 10^{18}$ no of e are less

$Q = -8 \times 10^{-20} C$

$Q = -0.8 \times 10^{-19} C$

$|Q| = ne$

$n = \dfrac{|Q|}{e} = \dfrac{0.8 \times 10^{-19}}{1.6 \times 10^{-19}} = 0.5$

Not possible.

Electric Charge

It is a property of matter which gives it the ability to repel or attract other charged substance placed in its vicinity.

Types of charges

Positive charge



Negative charge



- SI unit of charge is Coulomb.
- It is denoted by "C".
- Charge on a electron,

$$q_e = -1.6 \times 10^{-19}\ C$$

- Charge on a proton,

$$q_p = +1.6 \times 10^{-19}\ C$$

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Let's Solve



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Let's Solve

Q. If we remove 2 electrons from a Helium atom, how much charge will it have? (e = magnitude of charge on an electron)

A. +2e

B. -2e

C. -1e

D. +1e

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Let's Solve

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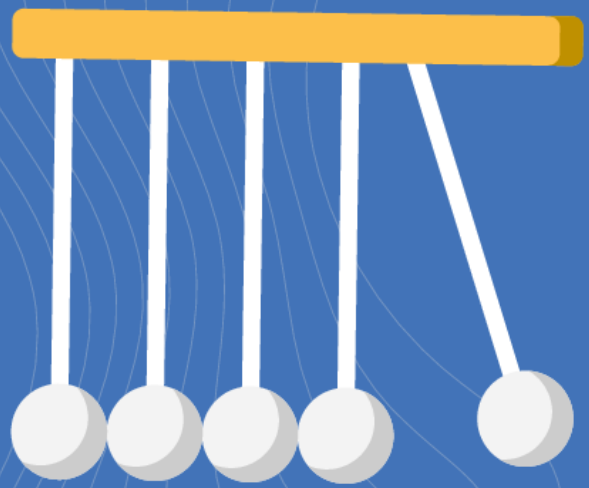
Let's Solve

Q. If we remove 2 electrons from a Helium atom, how much charge will it have? (e = magnitude of charge on an electron)

Solution:

Initially the whole atom is neutral. When we remove 2 electrons, it will have deficiency of 2 electrons which have -2e charge which will make the atom charged by +2e.

Quiz



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Quiz

Chat Off

Q. Three neutral balls A,B,C (identical in all aspects) are taken and then balls A and B are charged positively and negatively respectively. Arrange the balls in decreasing order of their final masses.

A. A > B > C

B. C > A > B

C. B > C > A

D. B > A > C

A	B	C
+ve	−ve	0
remove e	add e	
mass ↓	mass ↑	mass = same

B > C > A

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Quiz

Chat On

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B. C > A > B

C. B > C > A

D. B > A > C

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Quiz

Chat On

Q. Three neutral balls A,B,C (identical in all aspects) are taken and then balls A and B are charged positively and negatively respectively. Arrange the balls in decreasing order of their final masses.

Solution:

Ball A is positively charged that means some electrons are taken away from this ball. So, Its mass will decrease.

Ball B is negatively charged that means some electrons are added in this ball. So, its mass will increase.

Ball C is still neutral. That means no decrement or increment in its mass.

As all the balls are initially identical. So, B > C > A

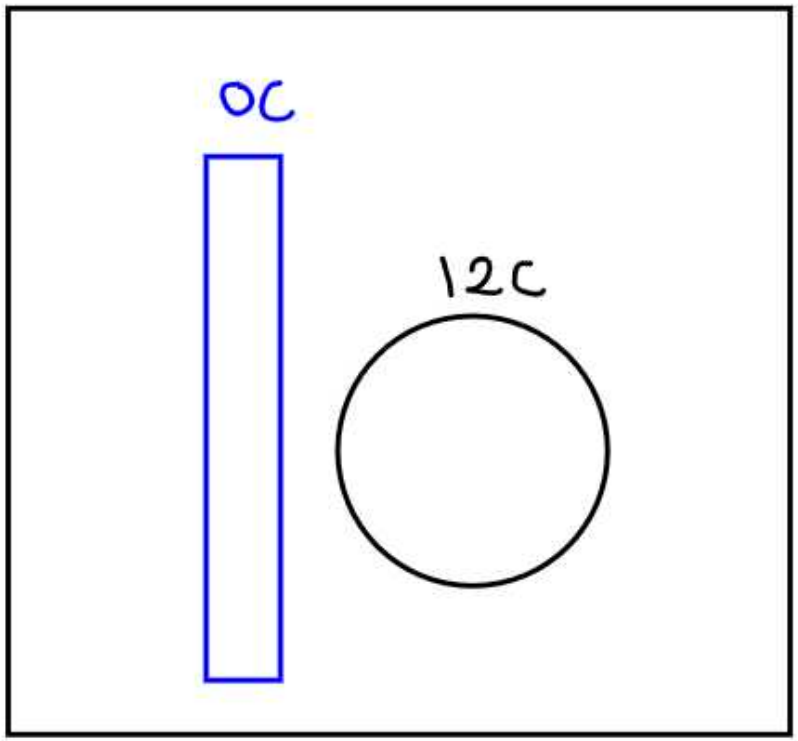
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Properties of Electric Charge

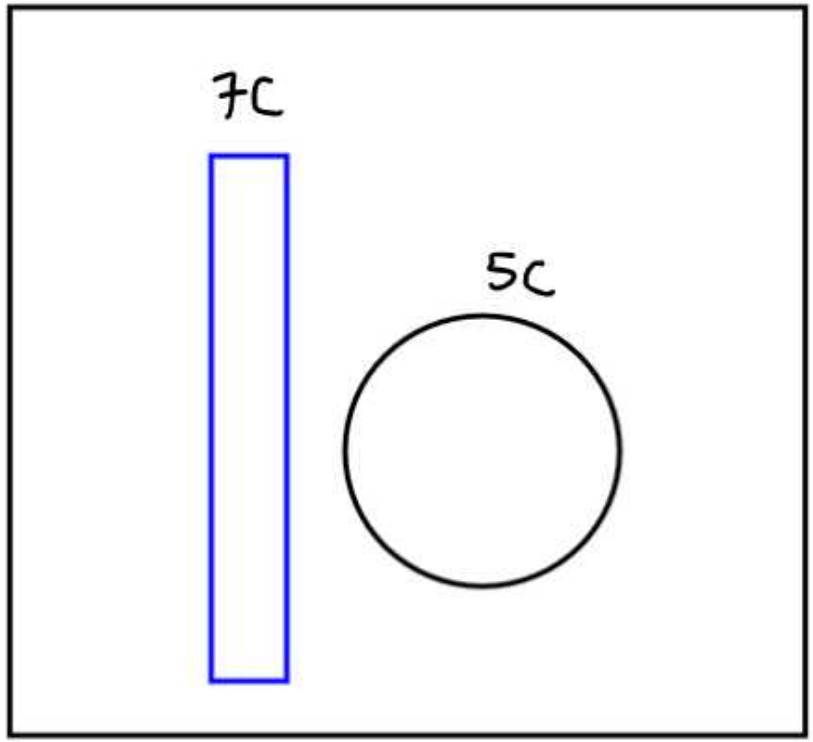
- Additivity of charges
- Quantization
- Conservation of charge

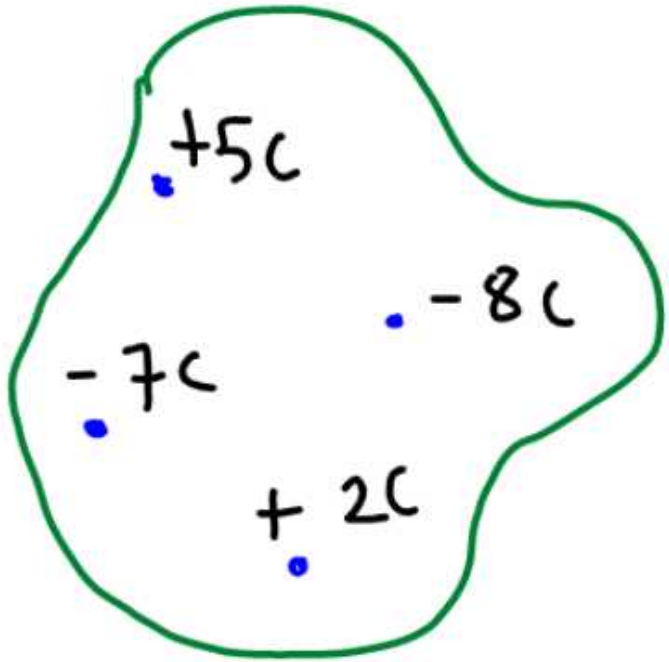
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Isolated.



Isolated.





Q. Find the total charge of the system.

$$Q = +5 - 7 - 8 + 2 = -8C$$

Additivity of charges

- The total charge of the system is obtained simply by adding all the charges algebraically.
- If a system contains n charges q_1, q_2, q_3, ..., q_n, then

  total charge of the system = $q_1 + q_2 + q_3 + ... + q_n$.
- Charge has magnitude but no direction, similar to mass.
- However, mass of a body is always positive whereas a charge can be either positive or negative.

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Quantization

- The smallest charge that can be added to or removed from a body is the charge of an electron (e).
- If charge on a body is Q,then

$$Q = ne$$

Where n = integer

$$e = 1.6 \times 10^{-19}\ C$$

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Let's Solve



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Let's Solve

Q.What will be the total charge of a system containing five charges +2 C, +3 C, −3 C, +5 C and −6 C?

A +19 C

B -9 C

C -1 C

D +1 C

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Let's Solve

Q.What will be the total charge of a system containing five charges +2 C, +3 C, −3 C, +5 C and −6 C?

A. +19 C

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C. -1 C

D. +1 C

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Let's Solve

Q.What will be the total charge of a system containing five charges +2 C, +3 C, −3 C, +5 C and −6 C?

Solution:

Total charge $(Q) = q_1 + q_2 + q_3 + q_4 + q_5$

$\Rightarrow Q = (+2) + (+3) + (-3) + (+5) + (-6)$

$\Rightarrow Q = +1\ C$

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Let's Solve



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Let's Solve

Q. What will be the number of electrons in 1 Coulomb charge?

A. 6.25×10^{18}

B. 6.25×10^{19}

C. 1.6×10^{19}

D. 1.6×10^{18}

$|Q| = ne$

$1 = n \times 1.6 \times 10^{-19}$

$\frac{1 \times 10^{19}}{1.6} = n$

$\frac{10 \times 10^{19}}{16} = n$

$0.625 \times 10^{19} = n$

$n = 6.25 \times 10^{18}$

$n = 6250000000 \ldots\ldots -$

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Let's Solve

Q. What will be the number of electrons in 1 Coulomb charge?

A 6.25×10^{18}

B 6.25×10^{19}

C 1.6×10^{19}

D 1.6×10^{18}

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Let's Solve

Q. What will be the number of electrons in 1 Coulomb charge?

Solution:

$$Q = ne$$

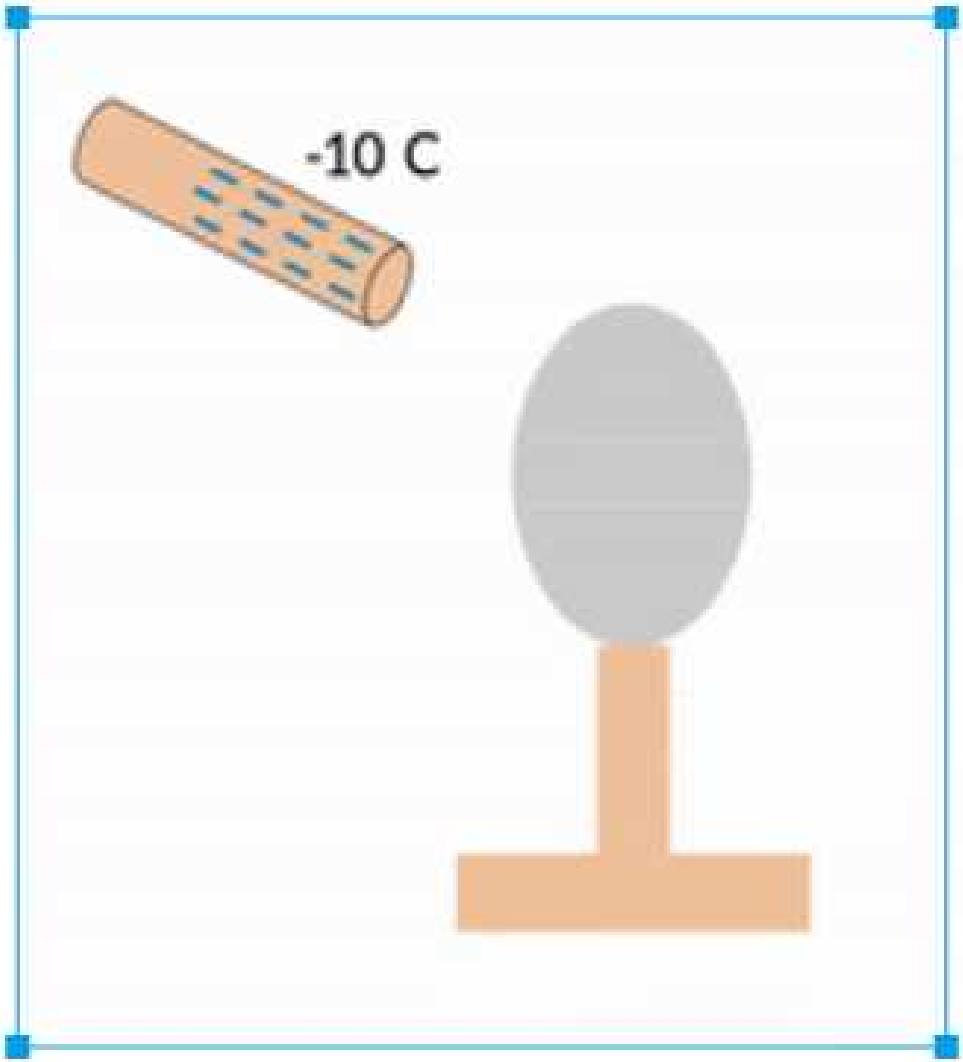
$$n = \frac{Q}{e} = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18}$$

That means one coulomb charge has 6.25×10^{18} electrons.

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Conservation of Charge

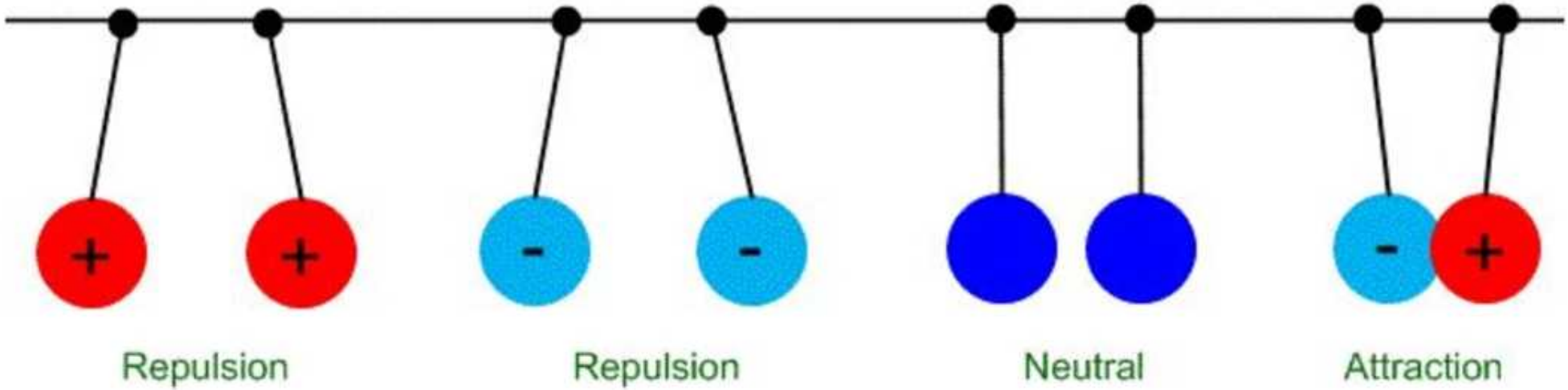
For an isolated system, charge can neither be created nor be destroyed i.e. the net charge of an isolated system remains constant.



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Auxiliary Properties of charges

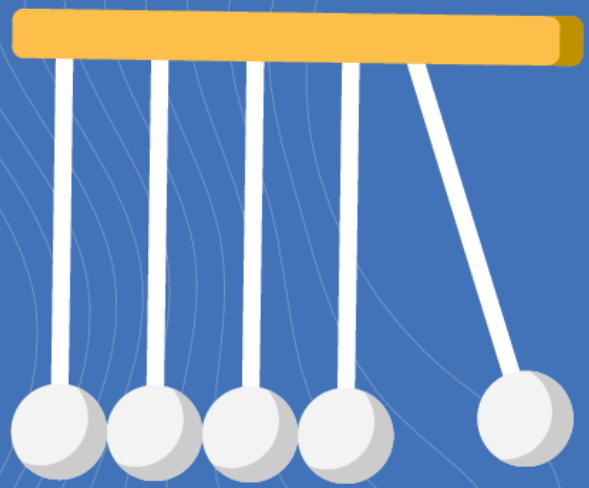
- Charge is always associated with matter.
- Like charges repel and unlike charges attract each other.



- Charge on a particle is independent of the state of rest or motion of the particle.

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Quiz



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Quiz

Chat Off

Q. We wish to give a charge of 0.6×10^{-19} C to a ball. Is it possible?

A Yes

B No

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Quiz

Chat On

Q. We wish to give a charge of 0.6×10^{-19} C to a ball. Is it possible?

A Yes

B No

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Quiz

Chat On

Q. We wish to give a charge of 0.6 x 10^{-19} C to a ball. Is it possible?

Solution:

$$Q = ne$$

$$\Rightarrow n = \frac{Q}{e}$$

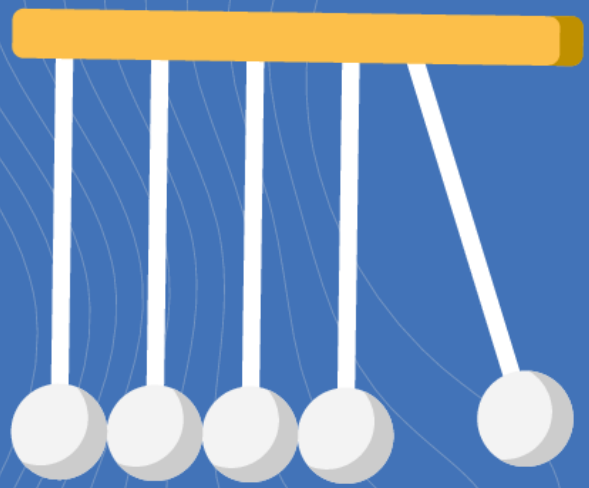
$$\Rightarrow n = \frac{0.6 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$$\Rightarrow n = 0.375$$

Because, n is not an integer. So it is not possible to give 0.6 x 10^{-19} C of charge to a ball.

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Quiz



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Quiz

Chat Off

Q. A conductor has a positive charge of 14.4×10^{-19}C .The conductor has .....electrons as compared to neutral conductor.

A 9 more

B 9 less

C 27 more

D 27 less

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Quiz

Chat On

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B 9 less

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D 27 less

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Quiz

Chat On

Q. A conductor has a positive charge of 14.4×10^{-19}C .The conductor has .....electrons as compared to neutral conductor.

Solution:

Charge on the conductor (Q) = $+(14.4\times10^{-19})$C

As the conductor is positively charged, so it is short of electrons.

By quantization of electric charges,

$Q = ne$

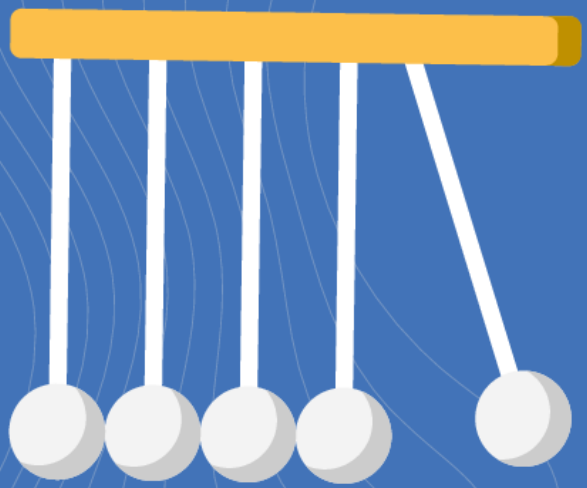
$\Rightarrow 14.4\times10^{-19} = n(1.6\times10^{-19})$

$\Rightarrow n=9$

Thus number of deficient electrons are 9.

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Quiz



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Quiz

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Q. Two charged bodies P and Q attract each other. Now bodies Q and R repel each other. If P and R are held close together, they will

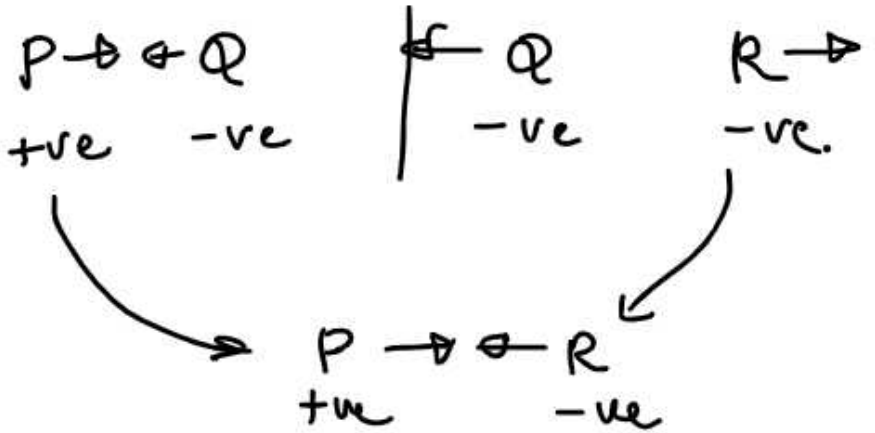
A Repel

B Attract ✓

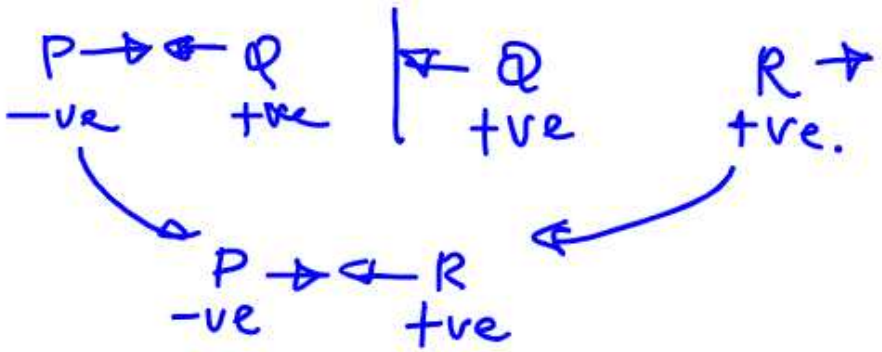
C Not affect each other

D Information insufficient

Case I



Case II



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Quiz

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Q. Two charged bodies P and Q attract each other. Now bodies Q and R repel each other. If P and R are held close together, they will

A Repel

B Attract

C Not affect each other

D Information insufficient

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Quiz
Chat On

Q. Two charged bodies P and Q attract each other. Now bodies Q and R repel each other. If P and R are held close together, they will

Solution:

Body R is charged as only like charges can repel each other. Let us consider all the possible cases

Case I	Case II
If P and Q attract each other, + P − Q	If P and Q attract each other, − P + Q
Q and R repel each other. − Q − R	Q and R repel each other + Q + R

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Quiz

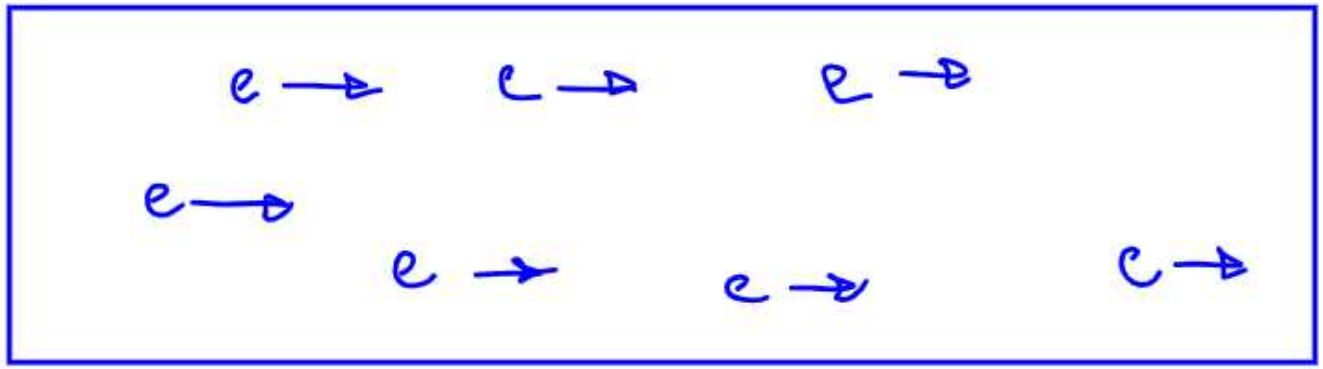
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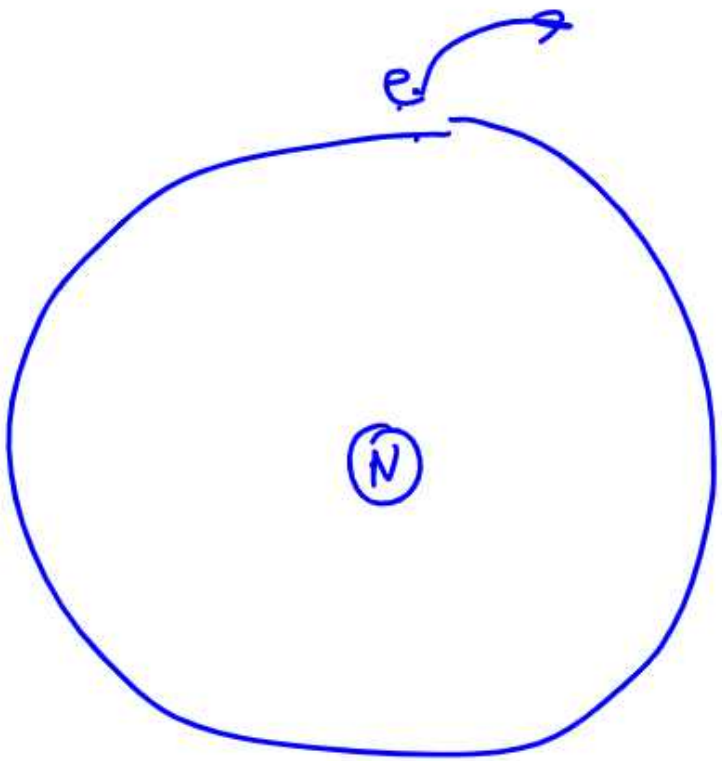
Q. Two charged bodies P and Q attract each other. Now bodies Q and R repel each other. If P and R are held close together, they will

Solution:

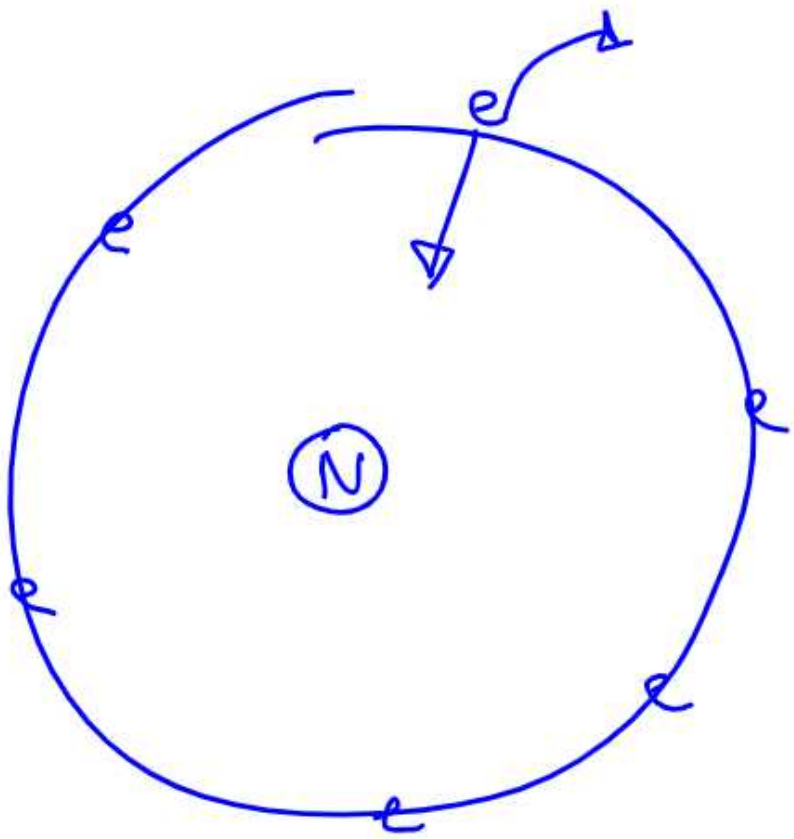
In both cases, P and R have unlike charges.
Hence, if the charges P and R are held close together, they will attract each other.

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Metals



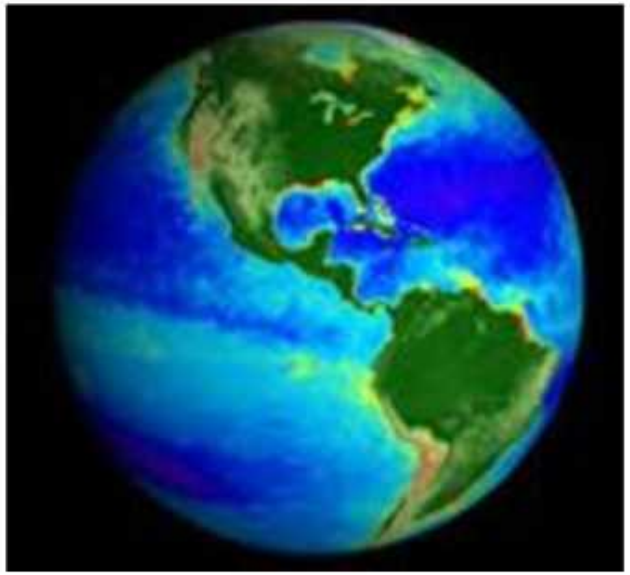
Insulator

Conductors

Materials that allow the flow of charges through them. Their conducting property is attributed to fact that they possess mobile charged particles in them.



Metals



Earth

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Insulators/Dielectric/Non-conductors

Materials that do not allow the flow of charges through them.

They lack mobile charged particles in them.



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Methods of charging a body

Charging a body is addition or removal of electrons. Charging is done by several methods.

- By Friction (Rubbing) → Only for Non conductor
- By Conduction (Touching) → only for conductors.
- By Induction → only for conductors.

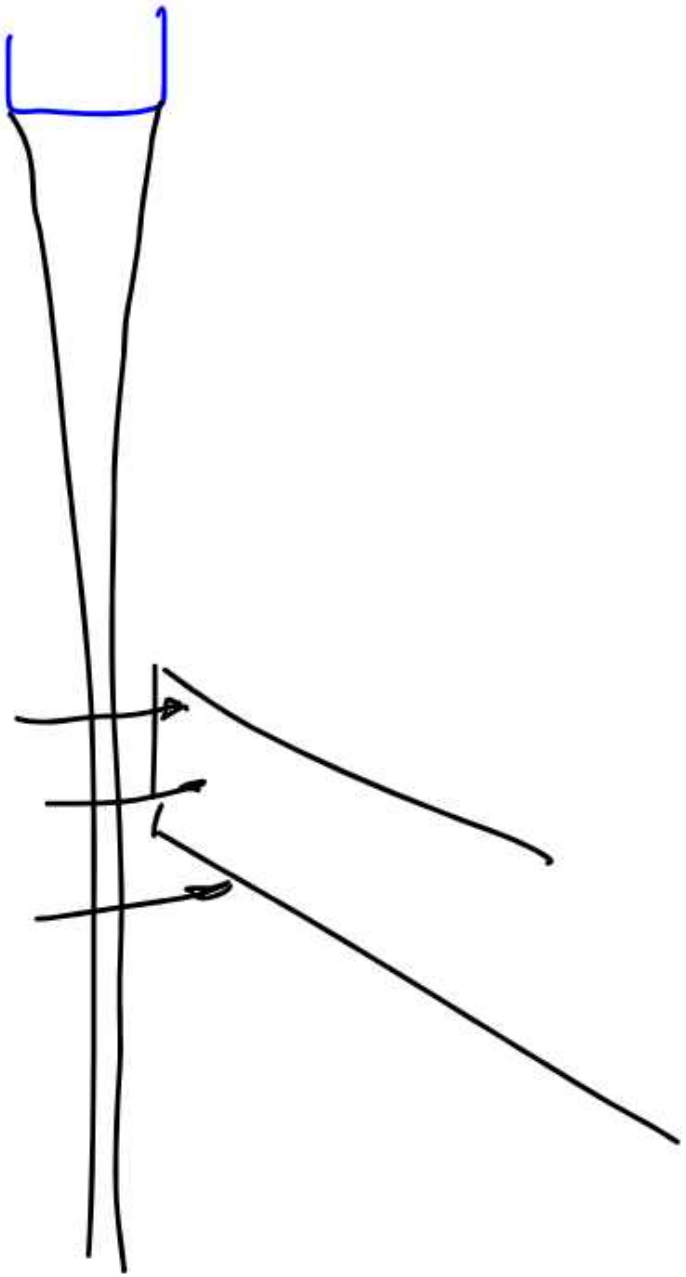
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Charging an Insulator

Insulators can be charged using friction. Example, Glass rod and silk cloth.



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Conductors : Important properties

- Two conductors kept in contact behave as one.
- The charge given to a conductor distributes itself on the outer surface of the conductor only. The distribution may or may not be uniform.
- There are two more common methods of charging of a conductor:

  I. Conduction

  II. Induction

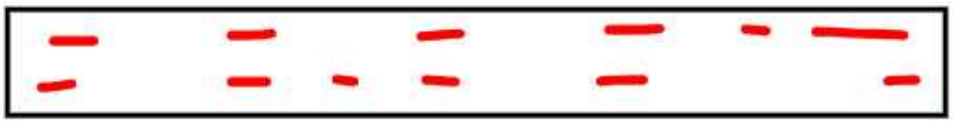
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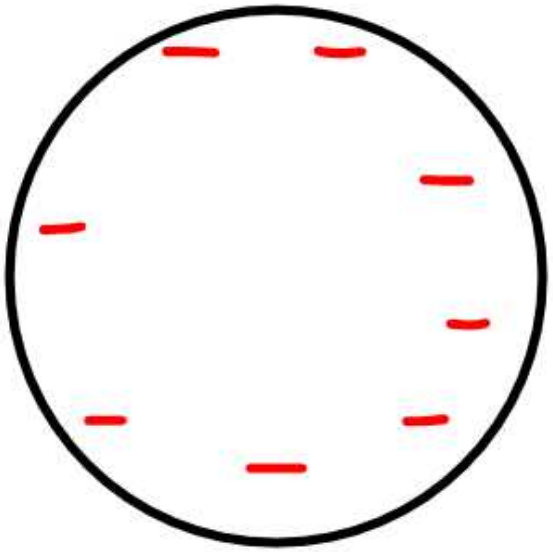
Charging by Conduction



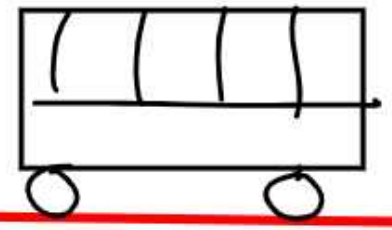
In charging by conduction both objects get the same type of charge.

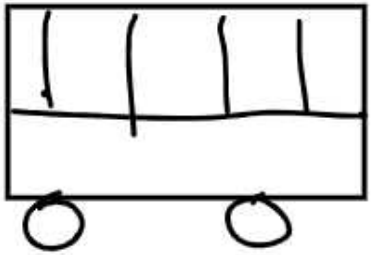
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Charging by Induction

The process of redistribution of charge on the surface of a conductor when another charged object is brought near the conductor, is called induction.



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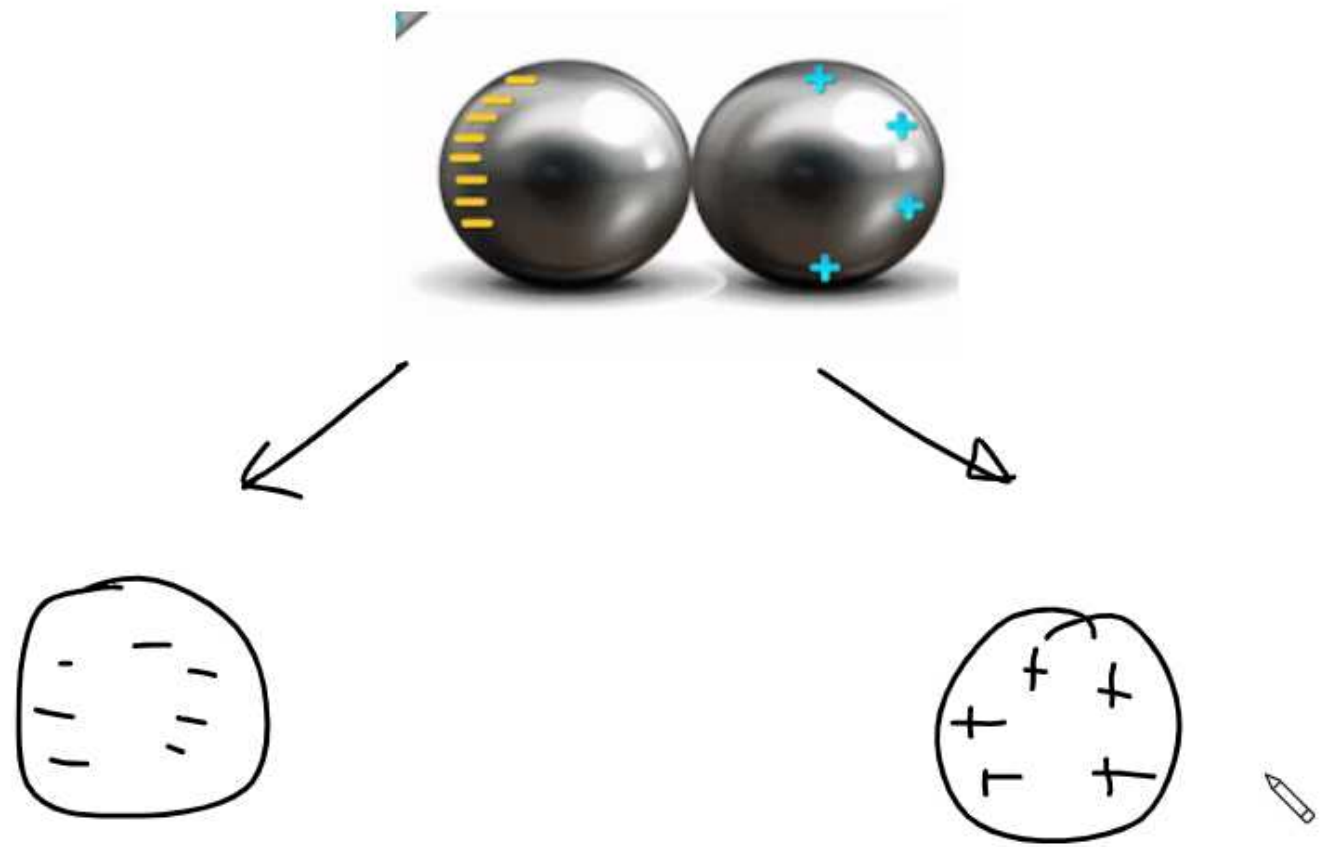
Charging by Induction

The process of redistribution of charge on the surface of a conductor when another charged object is brought near the conductor, is called induction.

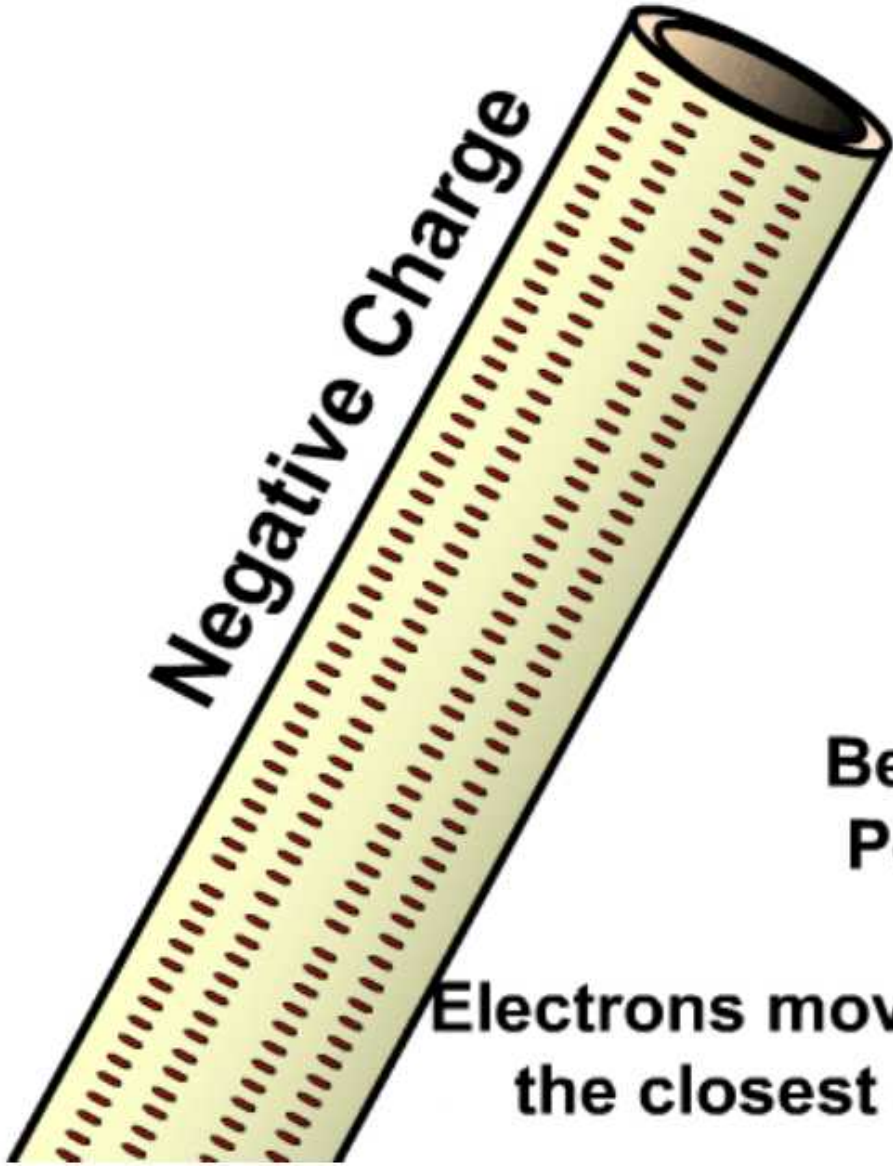
The other charged object may/may not be a conductor



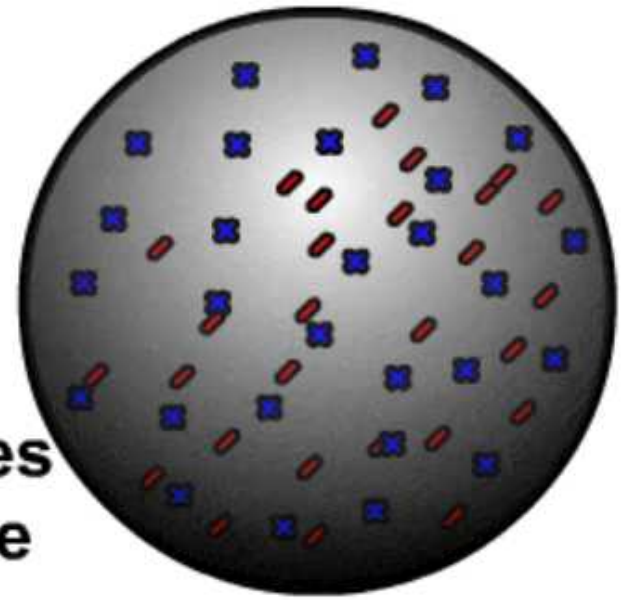
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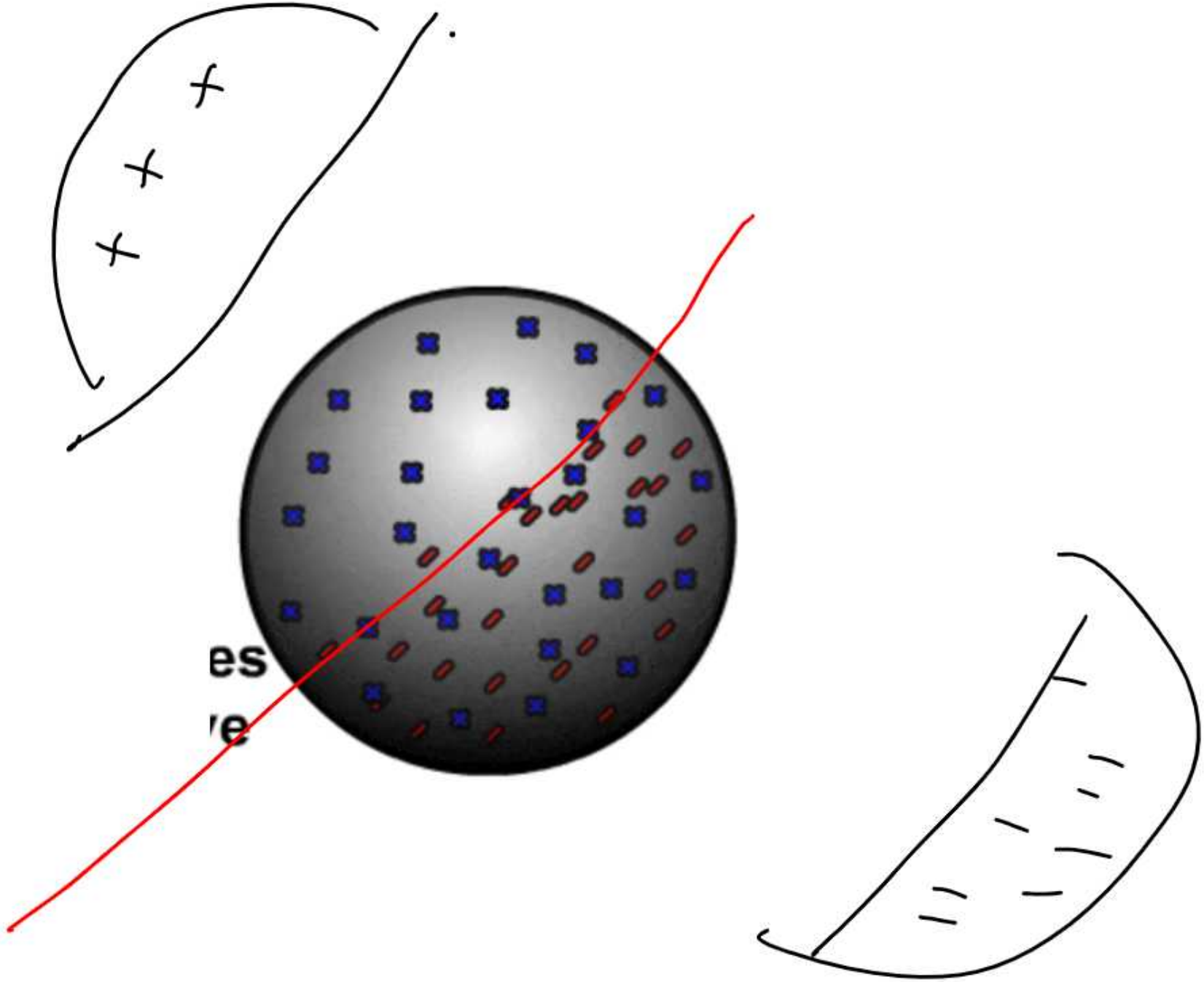
Induction



Becomes Positive



Electrons move within the object making the closest side oppositely charged



Methods of charging a body



Interactivity Pod

On charging by conduction, mass of a body may

A increase

B decrease

C Remain same

D Increase or decrease

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Methods of charging a body



Interactivity Pod

On charging by conduction, mass of a body may

A increase

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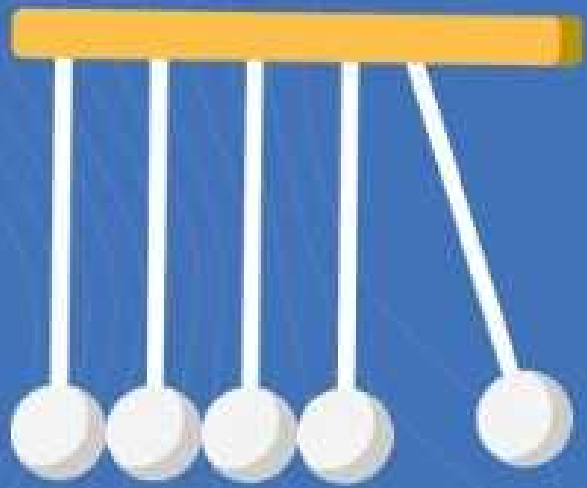
D Increase or decrease

Solution:

A body acquiring positive charge is actually losing electrons and hence its mass will become less. Similarly when it acquires negative charge, it accepts electrons and its mass increases.

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Quiz



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Quiz

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Q. On rubbing, when one body gets positively charged and other negatively charged, the electrons transferred from positively charged body to negatively charged body are

A. valence electrons only

B. electrons of inner shells

C. both valence electrons and electrons of inner shell

D. yet to be established

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Quiz

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Q. On rubbing, when one body gets positively charged and other negatively charged, the electrons transferred from positively charged body to negatively charged body are

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B. electrons of inner shells

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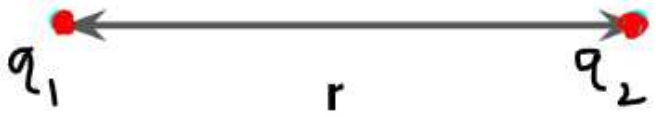
D. yet to be established

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Coulomb's Law

$F = \frac{G m_1 m_2}{r^2}$

$k = \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \ N m^2/C^2$



i.e. $F \propto q_1 q_2 / r^2$

$$F = k \frac{|q_1 \ q_2|}{r^2}$$

This is called Coulomb's law.

$$F = \frac{1}{4\pi\varepsilon_0} \frac{|q_1 q_2|}{r^2}$$

magnitude.

where, k = Coulomb's constant

In SI system $k = 9 \times 10^9 \ N \ m^2/C^2$

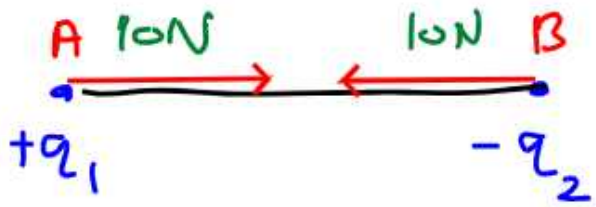
$k = 1/4\pi\varepsilon_0$

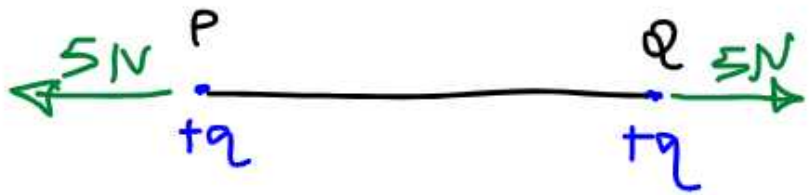
ε_0 = permittivity of free space

$\varepsilon_0 = 8.854 \times 10^{-12} \ C^2 \ N^{-1} m^{-2}$

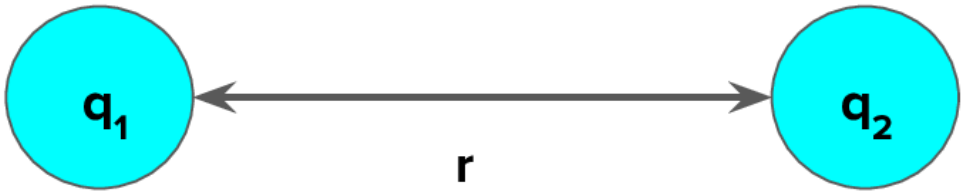
- This law is valid only for point charges.

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A
10N
10N
B
+q1
-q2


5N
P
Q
5N
+q
+q

Coulomb's Law



$$F = k \frac{|q_1 \ q_2|}{r^2}$$

If $q_1 = q_2 = 1$ C and r = 1 m,

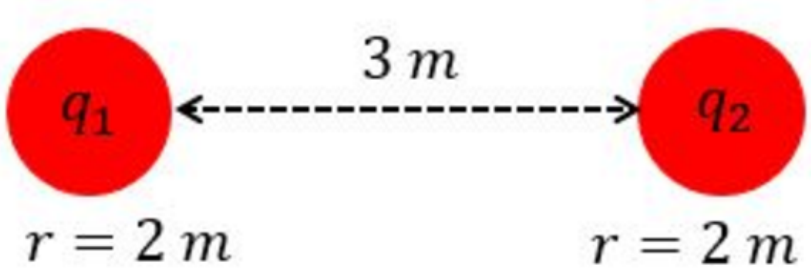
Then, F = k.

Coulomb's constant is numerically equal to the electrostatic force between two unit charges placed at unit distance from each other **in vacuum.**

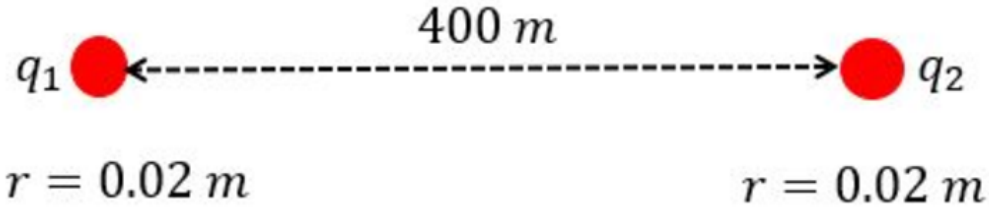
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Limitations of Coulomb's Law

- The charges are to be considered as point charges.



Coulomb's law is not applicable, as distance between charged spheres and radius are comparable.



Coulomb's law is applicable, as separation between charges is large as compared to their dimensions.

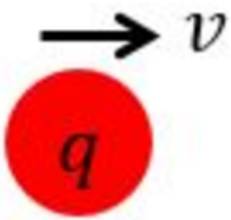
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Limitations of Coulomb's Law

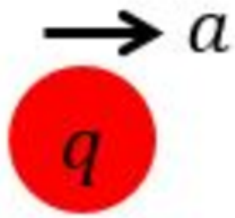
- Charges should be stationary.



If charge is at rest, then it produces only electric field.



If charge is moving with constant velocity, then both Electric field & Magnetic fields are produced by it. We will study about magnetic field in later chapters.



If charge is moving with constant acceleration, then it produces Electromagnetic waves . We will study about this in later chapters.

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Limitations of Coulomb's Law



Interactivity Pod

Two particles having charges 1 μC and q mC are separated by a distance of 3 m. If repulsive force on one due to other is 2 N. Find the value of q.

A 3

B 2

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Limitations of Coulomb's Law



Interactivity Pod

Two particles having charges 1 μC and q mC are separated by a distance of 3 m. If repulsive force on one due to other is 2 N. Find the value of q.

A 3

B 2

$$F = \frac{kq_1q_2}{r^2}$$

$$\Rightarrow 2 = \frac{9 \times 10^9 \times (1 \times 10^{-6}) \times (q \times 10^{-3})}{(3)^2}$$

$$\Rightarrow 18 = 9q$$

$$\Rightarrow q = 2$$

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Let's Solve



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Let's Solve

Q. Two particles having charges 3 μC and 2 mC are separated by a distance of 6 m. Find electrostatic force on one due to other?

A. 3 N, repulsive

B. 1.5 N, repulsive

C. 3 N, attractive

D. 1.5 N, attractive

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Let's Solve

Q. Two particles having charges 3 μC and 2 mC are separated by a distance of 6 m. Find electrostatic force on one due to other?

A. 3 N, repulsive

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C. 3 N, attractive

D. 1.5 N, attractive

Let's Solve

Q. Two particles having charges 3 μC and 2 mC are separated by a distance of 6 m. Find electrostatic force on one due to other?

Solution:

$$q_1 = 3\mu C = 3 \times 10^{-6} C$$

$$q_2 = 2\mu C = 2 \times 10^{-3} C$$

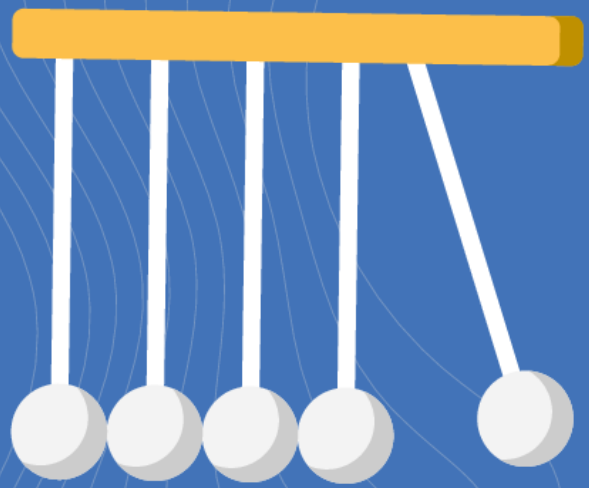
$$F = \frac{kq_1q_2}{r^2}$$

$$\Rightarrow F = \frac{9 \times 10^9 \times (3 \times 10^{-6}) \times (2 \times 10^{-3})}{(6)^2}$$

$$\Rightarrow F = 1.5\ N$$

Since charges are like , force will be repulsive.

Quiz



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Quiz

Chat Off

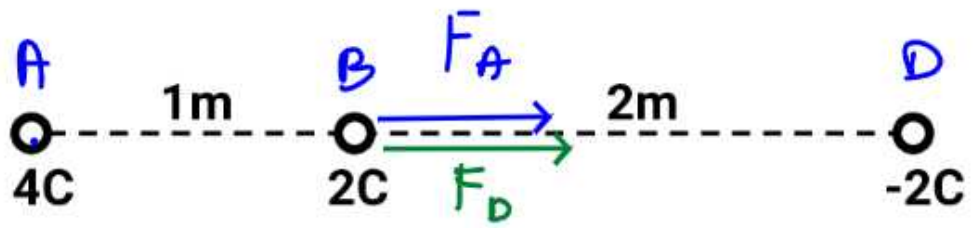
Q. Find the net force on 2C charge placed in the figure?

A. 0 N

B. 8.1×10^6 N

C. 81×10^9 N

D. 9 N



$$F_A = \frac{k(4)(2)}{1^2} = 8k$$

$$F_D = \frac{k(2)(2)}{2^2} = k$$

$$F_{Net} = F_A + F_D$$

$$= 8k + k$$

$$= 9k$$

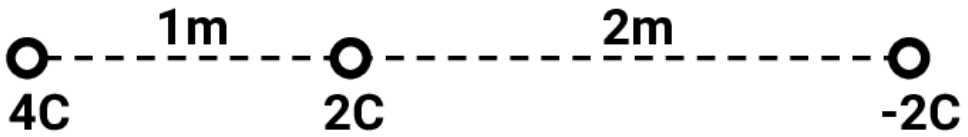
$$= 9 \times 9 \times 10^9 = 81 \times 10^9 \text{ N}$$

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Q. Find the net force on 2C charge placed in the figure?



A. 0 N

B. 8.1×10^6 N

C. 81×10^9 N

D. 9 N

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Q. Find the net force on 2C charge placed in the figure?

Solution: F_1 = Force on 2C due to 4C

F_2 = Force on 2C due to -2C

By principle of superposition; $F_{net} = F_1 + F_2$

$F_1 = k\frac{(4)(2)}{(1)^2}$ **Right Direction**

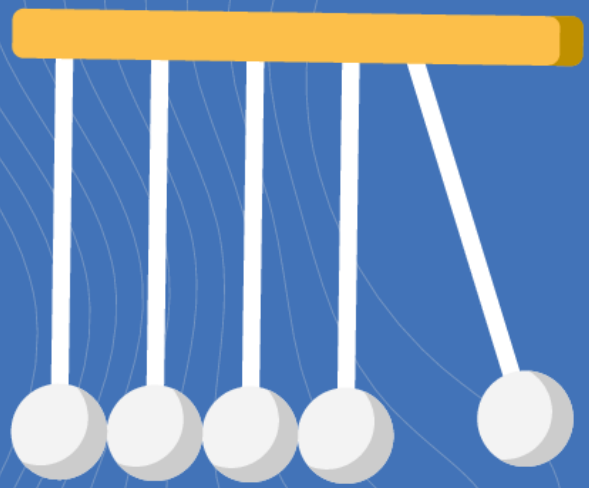
$F_2 = k\frac{(2)(2)}{(2)^2}$ **Right Direction**

$$F_{net} = 8k + 1k = 9k = 81 \times 10^9 \text{ N}$$

We do not consider sign of charge, if direction of force is seen through observation.

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Quiz



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Quiz

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Q. The electrostatic force on a small sphere of charge 0.4μC due to another small sphere of charge −0.8μC in air is 0.2N. What is the distance between the two spheres?

A 0.1 m

B 0.12 m

C 0.14 m

D 0.16 m

$$F = \frac{k|q_1 q_2|}{r^2}$$

$$0.2 = \frac{9\times10^9 \times 0.8\times10^{-6} \times 0.4\times10^{-6}}{r^2}$$

$$r^2 = 3^2 \times 10^8 \times 10 \times 1.6 \times (10^{-6})^2$$

$$r = 3\times10^4 \times 4 \times 10^{-6}$$

$$r = 12\times10^{-2}$$

$$r = 0.12\text{m}$$

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Quiz

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Quiz

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Q. The electrostatic force on a small sphere of charge 0.4μC due to another small sphere of charge −0.8μC in air is 0.2N. What is the distance between the two spheres?

Solution:

$$F = \frac{kq_1q_2}{r^2}$$

$$\Rightarrow 0.2 = \frac{9\times10^9\times(0.4\times10^{-6})(0.8\times10^{-6})}{r^2}$$

$$\Rightarrow r = 0.12\ m$$

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Summary+



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Summary

→ $q_e = -1.6 \times 10^{-19} C$ and $q_p = +1.6 \times 10^{-19} C$

→ **Properties of Electric Charge:**

- ***Additivity of Charges:*** The total charge of the system is obtained simply by adding all the charges algebraically.
- ***Quantization:*** The smallest charge that can be added to or removed from a body is the charge of the electron (e).
- ***Conservation of Charges:*** For an isolated system, charge can neither be created nor be destroyed i.e. the net charge of an isolated system remains constant.

→ **Auxiliary Properties of Charges:**

- Charge is always associated with matter.
- Like charges repel and unlike charges attract each other.
- Charge on a particle is independent of the state of rest or motion of the particle.

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Summary

- **Conductors:** Materials that allow the flow of charges through them. Their conducting property is attributed to fact that they possess mobile charged particles in them.

- **Insulators:** Materials that do not allow the flow of charges through them. They lack mobile charged particles in them.

- Charging a body is addition or removal of electrons. Charging is done by several methods.
  I. By Conduction (Touching)
  II. By Friction (Rubbing)
  III. By Induction

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Assignment No 1

PHYSICS-EL-01

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PHYSICS-EL-01

Topics Covered: *Properties of charges; Charging by contact; Charging by induction Attraction and repulsion between charges*

Total Count of Questions: *25*

Expected Time of Attempt by Students: *50 min*

1. Which of the following is not a property of electric charge?
   (a) Charges are additive
   (b) Charges are conserved
   (c) Charges exist only in multiples of 2e
   (d) Like charges repel each other
2. Which statement is true about electric charges?
   (a) They can be created or destroyed
   (b) Only negative charges can move
   (c) Electric charge is continuous
   (d) Total charge of an isolated system remains constant
3. Which particle carries a negative charge?
   (a) Proton (b) Neutron
   (c) Electron (d) Positron
4. When a glass rod is rubbed with silk, it becomes positively charged because
   (a) It gains protons (b) It loses protons
   (c) It gains electrons (d) It loses electrons
5. Which material is least likely to conduct electric charge?
   (a) Copper (b) Aluminum
   (c) Rubber (d) Silver
6. What happens when two like charges are brought near each other?
   (a) They attract (b) They repel
   (c) They merge (d) They neutralize
7. A charged object attracts a neutral object because of:
   (a) Magnetic force
   (b) Gravitational force
   (c) Electrostatic induction
   (d) Conduction
8. Which of the following is a method of charging a body?
   (a) Heating
   (b) Magnetic induction
   (c) Conduction
   (d) Vibration
9. In charging by contact, the final charges are:
   (a) Opposite
   (b) Equal and opposite
   (c) Unequal
   (d) Same sign
10. A body is said to be neutral when it has:
    (a) More protons than electrons
    (b) More electrons than protons
    (c) Equal number of protons and electrons
    (d) No protons or electrons
11. What is the SI unit of electric charge?
    (a) Ampere (b) Coulomb
    (c) Ohm (d) Volt
12. When a charged rod is brought near bits of paper, they get attracted. This happens due to:
    (a) Repulsion
    (b) Gravitational pull
    (c) Induced polarization
    (d) Magnetic attraction
13. A negatively charged object is repelled by another object. The second object is likely to be:
    (a) Uncharged
    (b) Negatively charged
    (c) Positively charged
    (d) Either positively charged or uncharged
14. What happens to the net charge in a closed system during any process?
    (a) It increases (b) It decreases
    (c) It remains constant (d) It becomes zero
15. Which of the following statements is correct?
    (a) Only charged bodies exert force
    (b) Uncharged bodies can exert electrostatic force
    (c) Neutral bodies cannot interact with charges
    (d) Charged bodies can attract or repel neutral bodies
16. What type of charge is acquired by a rubber rod when rubbed with fur?
    (a) Positive (b) Negative
    (c) Neutral (d) Double positive
17. Which process does not require physical contact?
    (a) Charging by friction
    (b) Charging by contact
    (c) Charging by induction
    (d) Discharging

18. When a charged body touches a neutral body, the neutral body gets charged by:
(a) Friction (b) Induction
(c) Contact (d) Radiation

19. Electric charge is:
(a) Continuous (b) Quantized
(c) Random (d) Imaginary

20. Which of the following supports the fact that electric charge is quantized?
(a) It flows continuously
(b) It is always positive
(c) It occurs in discrete units
(d) It has infinite values

21. If an object is grounded, what happens to excess charge on it?
(a) It gets amplified
(b) It disappears
(c) It gets transferred to Earth
(d) It remains on the object

22. A charged body can attract a neutral body due to:
(a) Repulsion
(b) Coulomb repulsion
(c) Force of conduction
(d) Induced charges

23. A conductor differs from an insulator in that a conductor:
(a) Has tightly bound electrons
(b) Has free electrons
(c) Cannot be charged
(d) Has no electrons

24. When an object is charged by induction, the charge it gets is:
(a) The same as the inducing charge
(b) Opposite to the inducing charge
(c) Neutral
(d) Variable

25. Two bodies with opposite charges are brought close to each other. They will:
(a) Repel (b) Attract
(c) Discharge each other (d) Disappear

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ANSWERS

1. (c)	2. (d)	3. (c)	4. (d)	5. (c)	6. (b)	7. (c)	8. (c)	9. (d)	10. (c)
11. (b)	12. (c)	13. (b)	14. (c)	15. (d)	16. (b)	17. (c)	18. (c)	19. (b)	20. (c)
21. (c)	22. (d)	23. (b)	24. (b)	25. (b)					

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